

Classification of Inflammatory Bowel Disease Using Microbial Taxonomic Abundance

- Final dataset: HMP2 / IBDMDB metagenomic taxonomic profiles
- Final prediction task: binary IBD vs nonIBD
- Contributor: Hamza Salahuddin

Problem and Motivation

- IBD includes Crohn's disease and ulcerative colitis
- Diagnosis can require invasive and expensive procedures
- Gut microbiome profiles may provide useful non-invasive screening support
- Project goal: test whether microbial taxonomic abundance can classify IBD vs nonIBD
- Clinical priority: missing true IBD cases is more costly than flagging extra follow-up cases

Milestone 1 Retrospective

- Original milestone direction used a smaller MLRepo-style microbiome dataset
- Recovered working subset: 74 merged samples and 3,677 microbial features
- Label counts: 59 Crohn's disease and 15 Healthy
- Milestone 1 established feasibility, basic EDA, and the project motivation
- Main limitation: too few samples for a strong final claim

Dataset Pivot After Milestone 2

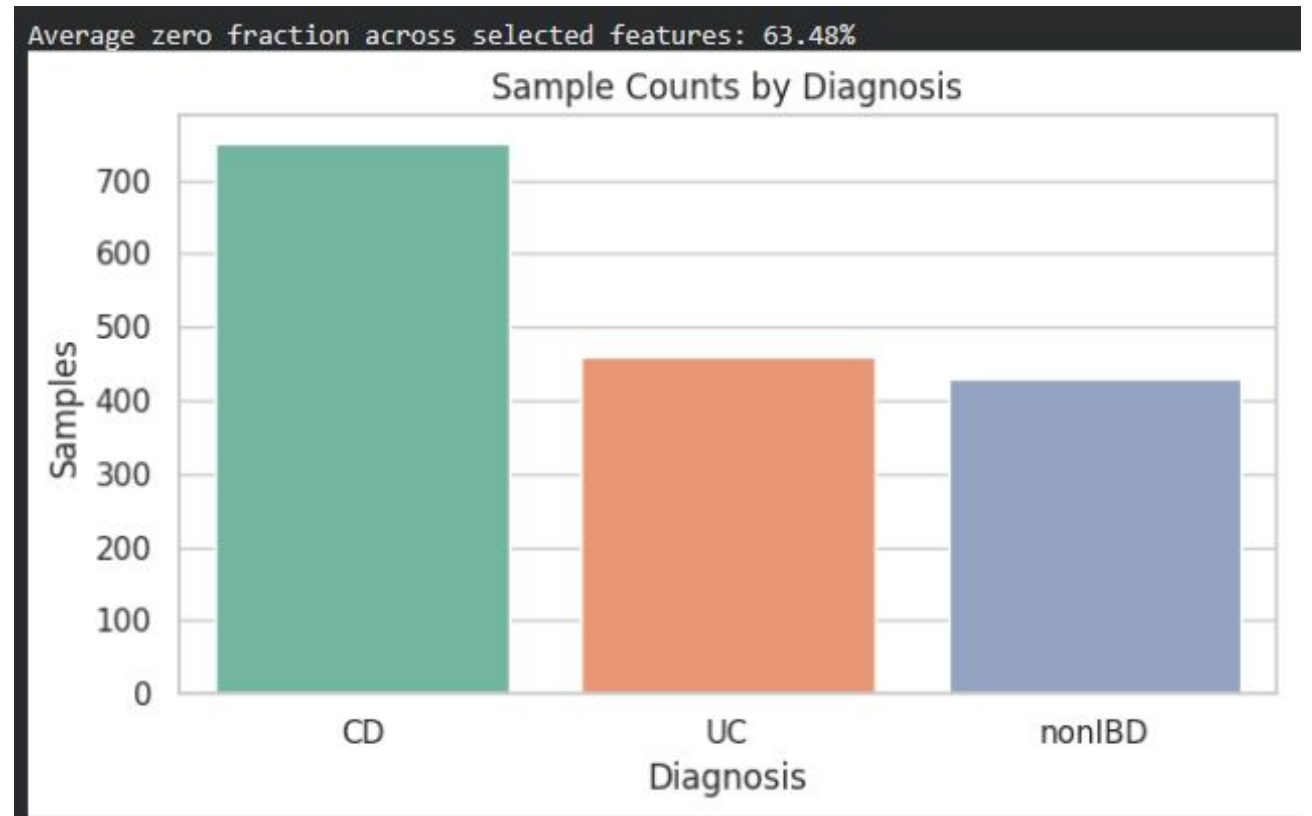
- Final dataset changed to HMP2 / IBDMDB for stronger evidence and better metadata
- Lloyd-Price et al. (2019) justified the HMP2 / IBDMDB dataset and showed strong IBD-associated microbial dysbiosis
- Source files used:
 - hmp2_metadata_2018-08-20.csv
 - taxonomic_profiles_3.tsv.gz
- Final merged modeling table: 1,638 samples from 130 participants
- Diagnosis counts: 750 CD, 459 UC, 429 nonIBD
- Final target: binary IBD vs nonIBD, with CD and UC collapsed into IBD

Preprocessing and Leakage Control

- Kept metagenomics samples only
- Morgan et al. (2012) and Gevers et al. (2014) supported the biological plausibility of microbiome-based IBD classification
- Merged metadata and taxonomy by sample ID
- Used genus-level features for interpretability
- Raw genus features: 187
- Selected features after training-only prevalence filtering: 78
- Applied log1p transformation to reduce skew and sparsity
- Used participant-aware train/test split to prevent leakage
- Train/test split: 1,333 train samples and 305 test samples
- Participant overlap between train and test: 0

Sample Counts by Diagnosis

- The diagnosis count chart shows the original three-label composition of the final HMP2 dataset
- Crohn's disease is the largest group, followed by ulcerative colitis, then nonIBD controls

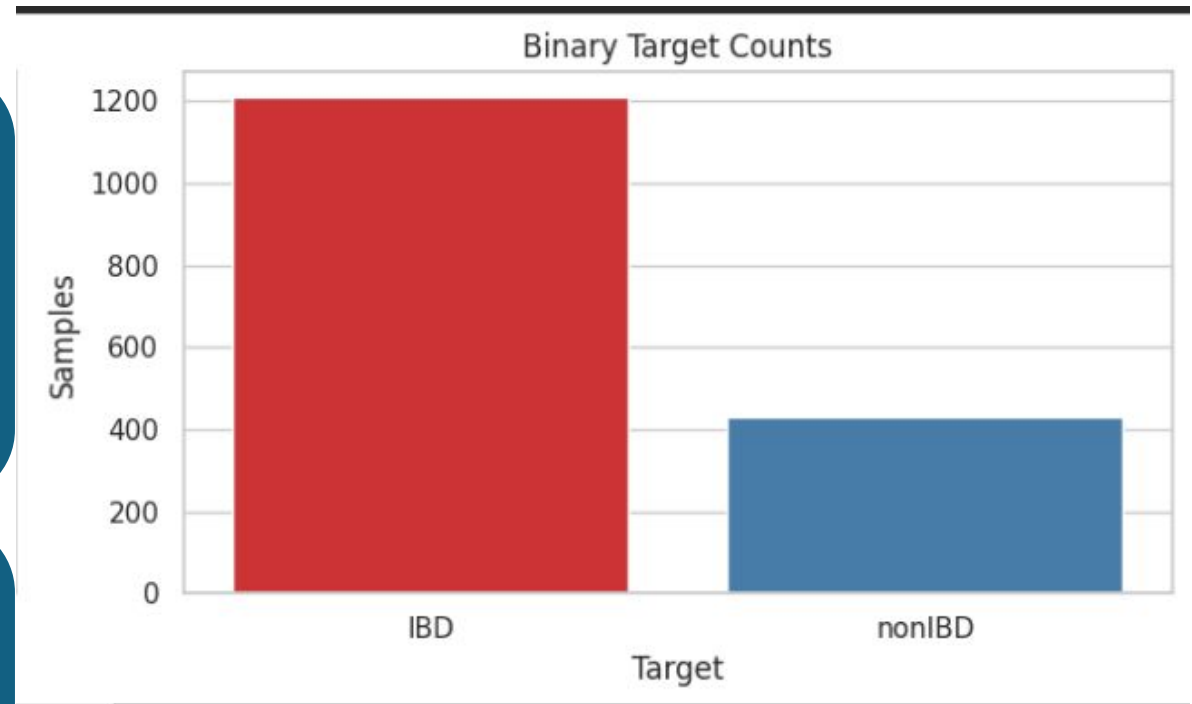


This chart shows why the raw data are naturally imbalanced before collapsing to the binary target

Binary Target Counts

The binary target chart shows the actual prediction task used for modeling: IBD vs nonIBD

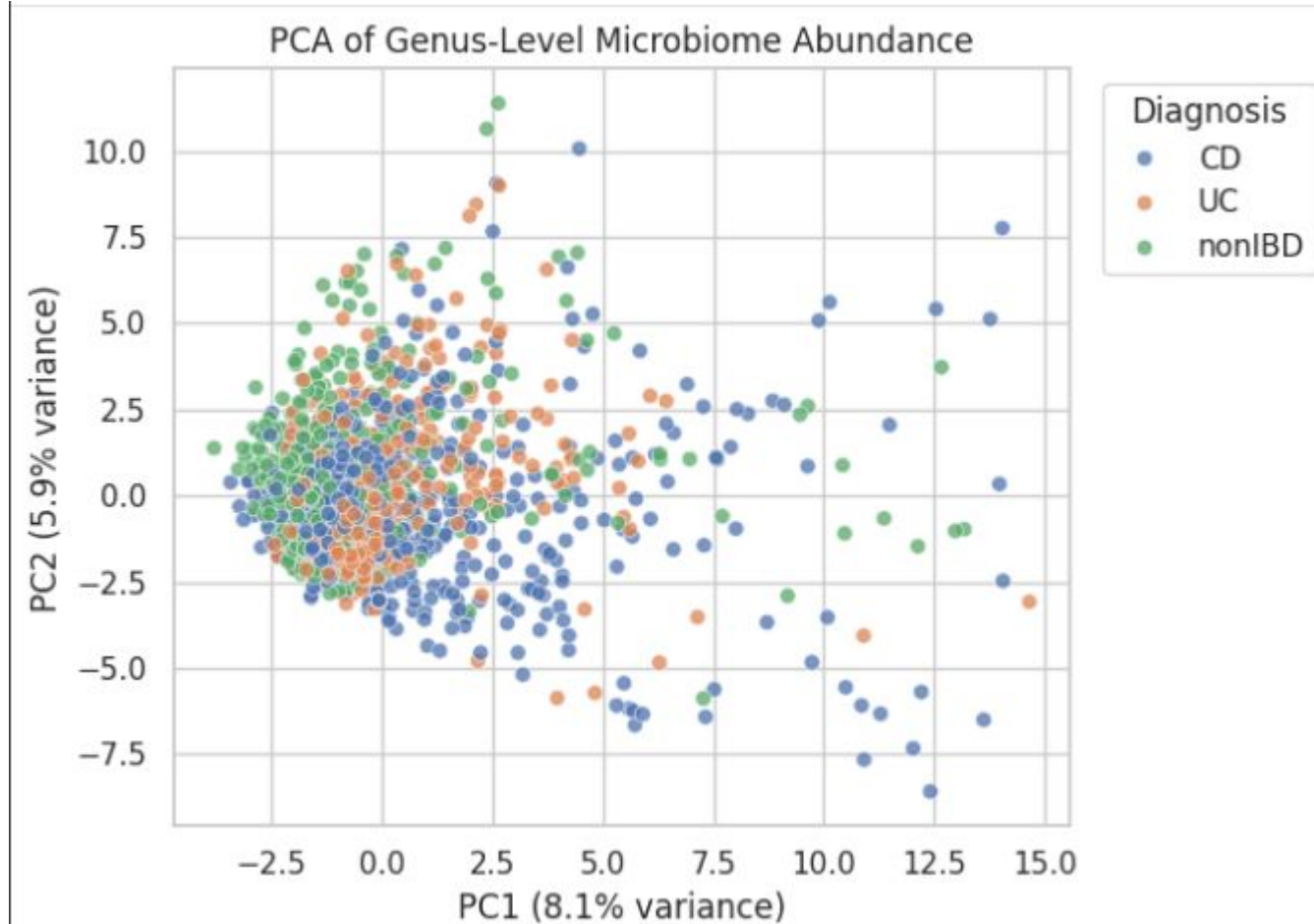
CD and UC were combined into the positive class IBD, while nonIBD remained the negative class



The chart confirms that IBD is the majority class, which is why accuracy alone is not enough for evaluation

PCA of Genus-Level Microbiome Abundance

- The groups show overlap, which means the classification problem is not trivially separable by eye
- Even with overlap, the spread suggests there is still structure in the microbiome data that supervised models can learn from



The PCA scatter plot projects high-dimensional genus abundance data into two principal components for visualization

Model Comparison Table

...	model	accuracy	balanced_accuracy	precision	recall	specificity	f1	roc_auc	pr_auc	tn	fp	fn	tp	
5	Gradient Boosting	0.800	0.600	0.871	0.895	0.306	0.882	0.763	0.944	15	34	27	229	
2	Random Forest	0.784	0.533	0.849	0.902	0.163	0.875	0.706	0.913	8	41	25	231	
4	SVM (RBF)	0.728	0.640	0.891	0.770	0.510	0.826	0.689	0.918	25	24	59	197	
3	Extra Trees	0.797	0.615	0.876	0.883	0.347	0.879	0.681	0.898	17	32	30	226	
1	Logistic Regression	0.603	0.582	0.877	0.613	0.551	0.722	0.656	0.880	27	22	99	157	
0	Majority baseline	0.839	0.500	0.839	1.000	0.000	0.913	0.500	0.839	0	49	0	256	
6	KNN	0.636	0.453	0.822	0.723	0.184	0.769	0.475	0.842	9	40	71	185	

The model comparison table evaluates each classifier on the same held-out test set

Gradient Boosting achieved the best ROC-AUC, while the majority baseline showed why high raw accuracy can be misleading in imbalanced data

This table supports choosing the final model based on recall, specificity, balanced accuracy, and ranking performance rather than accuracy alone

Threshold Tuning Summary

Best model by ROC-AUC: Gradient Boosting
Selected threshold from validation data: 0.670

	threshold	balanced_accuracy	f1	recall	specificity
124	0.670	0.670	0.740	0.636	0.703
128	0.690	0.668	0.721	0.606	0.730
113	0.615	0.666	0.775	0.697	0.635
129	0.695	0.666	0.718	0.602	0.730
125	0.675	0.666	0.735	0.629	0.703
114	0.620	0.664	0.772	0.693	0.635
117	0.635	0.663	0.763	0.678	0.649
123	0.665	0.663	0.738	0.636	0.689
130	0.700	0.662	0.712	0.595	0.730
126	0.680	0.662	0.729	0.621	0.703

The threshold tuning table shows how changing the classification cutoff changes recall and specificity

The default threshold favored very high IBD recall but weak nonIBD specificity

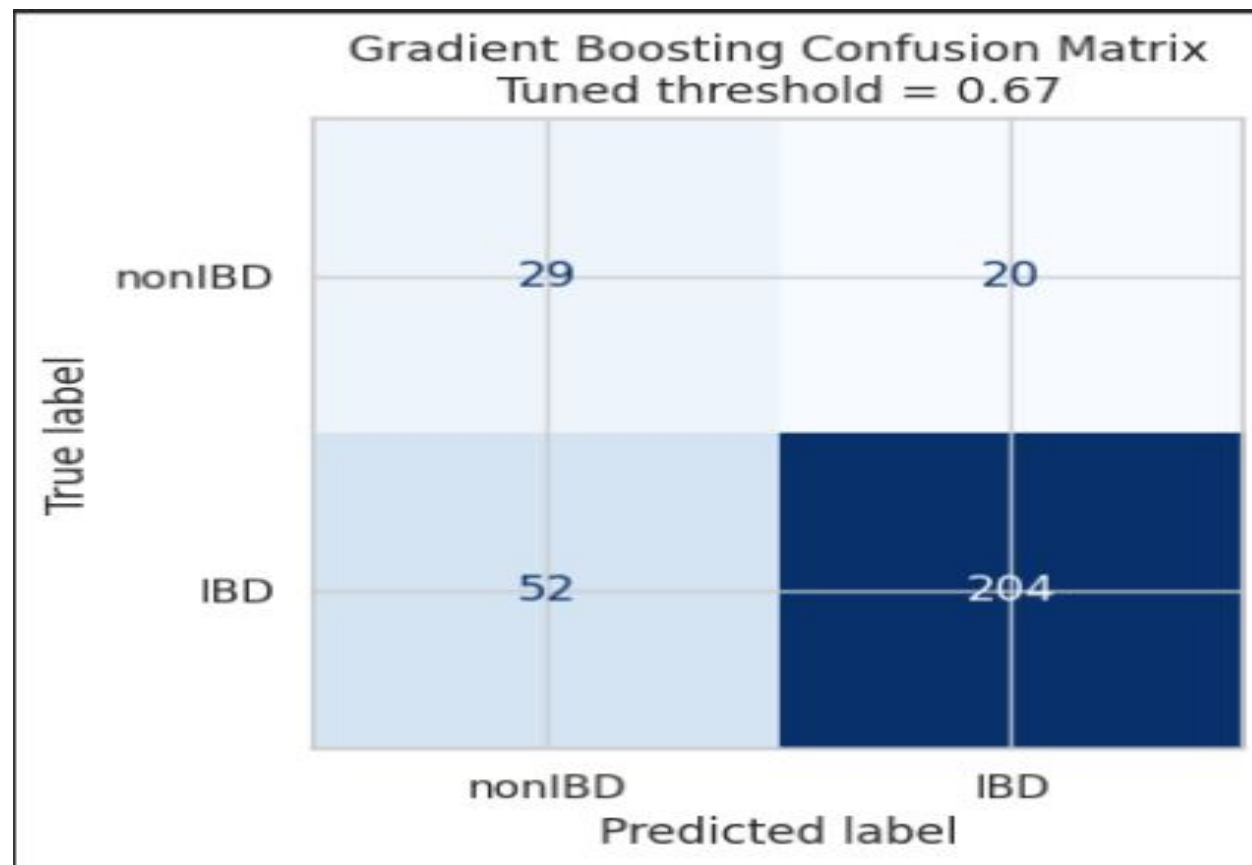
A tuned threshold of 0.67 produced a more balanced tradeoff, making the final model more clinically interpretable

	model	accuracy	balanced_accuracy	precision	recall_sensitivity	specificity	f1	roc_auc	pr_auc	tn	fp	fn	tp
0	Gradient Boosting default threshold	0.800	0.600	0.871	0.895	0.306	0.882	0.763	0.944	15	34	27	229
1	Gradient Boosting tuned threshold	0.764	0.694	0.911	0.797	0.592	0.850	0.763	0.944	29	20	52	204

Confusion Matrix with Tuned Threshold

It records 29 true negatives, 20 false positives, 52 false negatives, and 204 true positives

This figure gives the clearest single summary of the model's real classification behavior after threshold tuning



The confusion matrix shows the final tuned Gradient Boosting predictions on the untouched test set

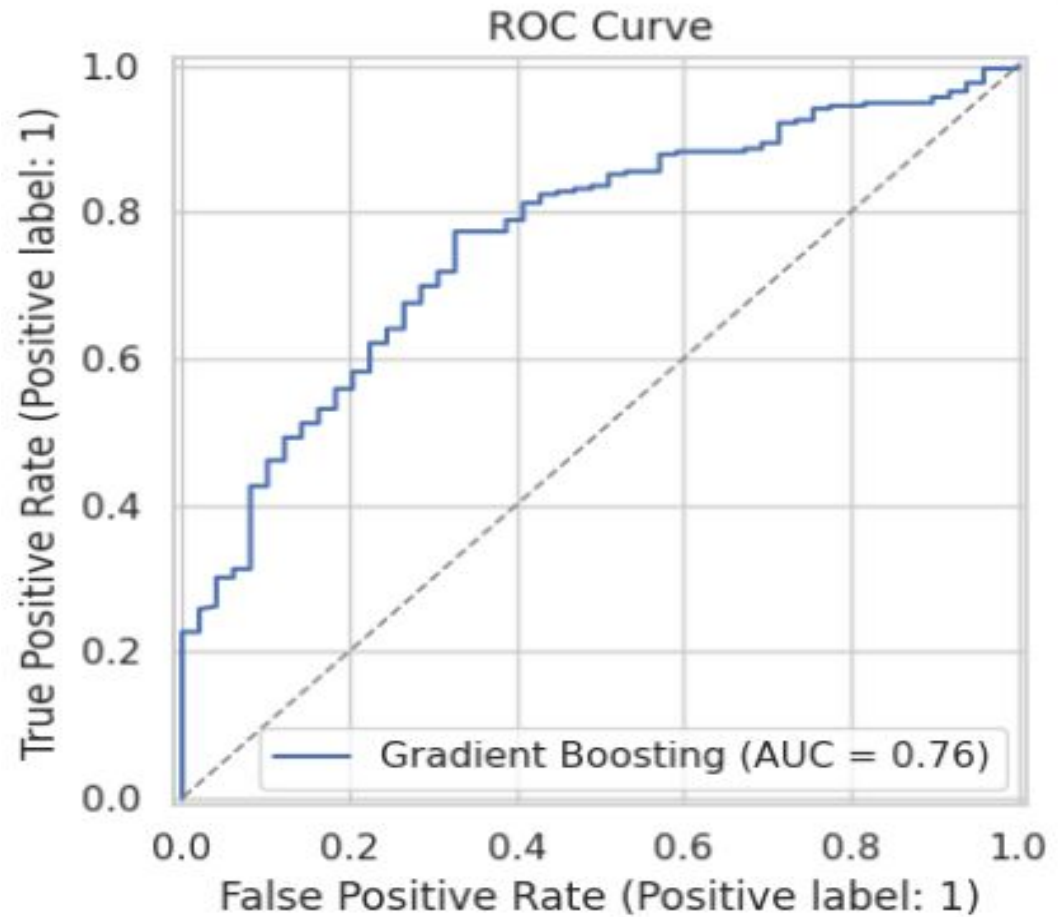
ROC Curve



The Gradient Boosting model achieved an ROC-AUC of about 0.76, which indicates meaningful discrimination above random guessing



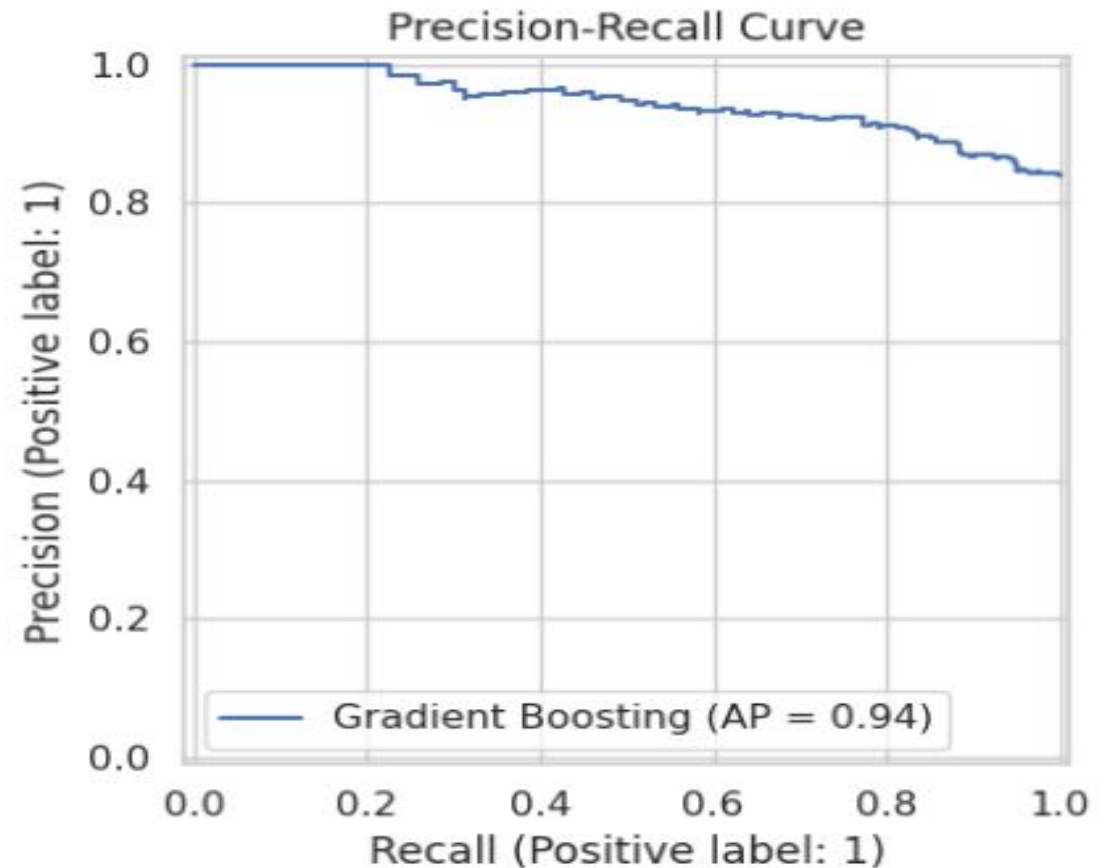
This curve is useful because it evaluates ranking quality independently of the final chosen threshold



The ROC curve shows the tradeoff between true positive rate and false positive rate across all possible thresholds

Precision-Recall Curve

- It is especially useful in imbalanced classification because it focuses on positive-class detection quality
- The high average precision indicates that the model maintains strong precision across a broad range of recall values

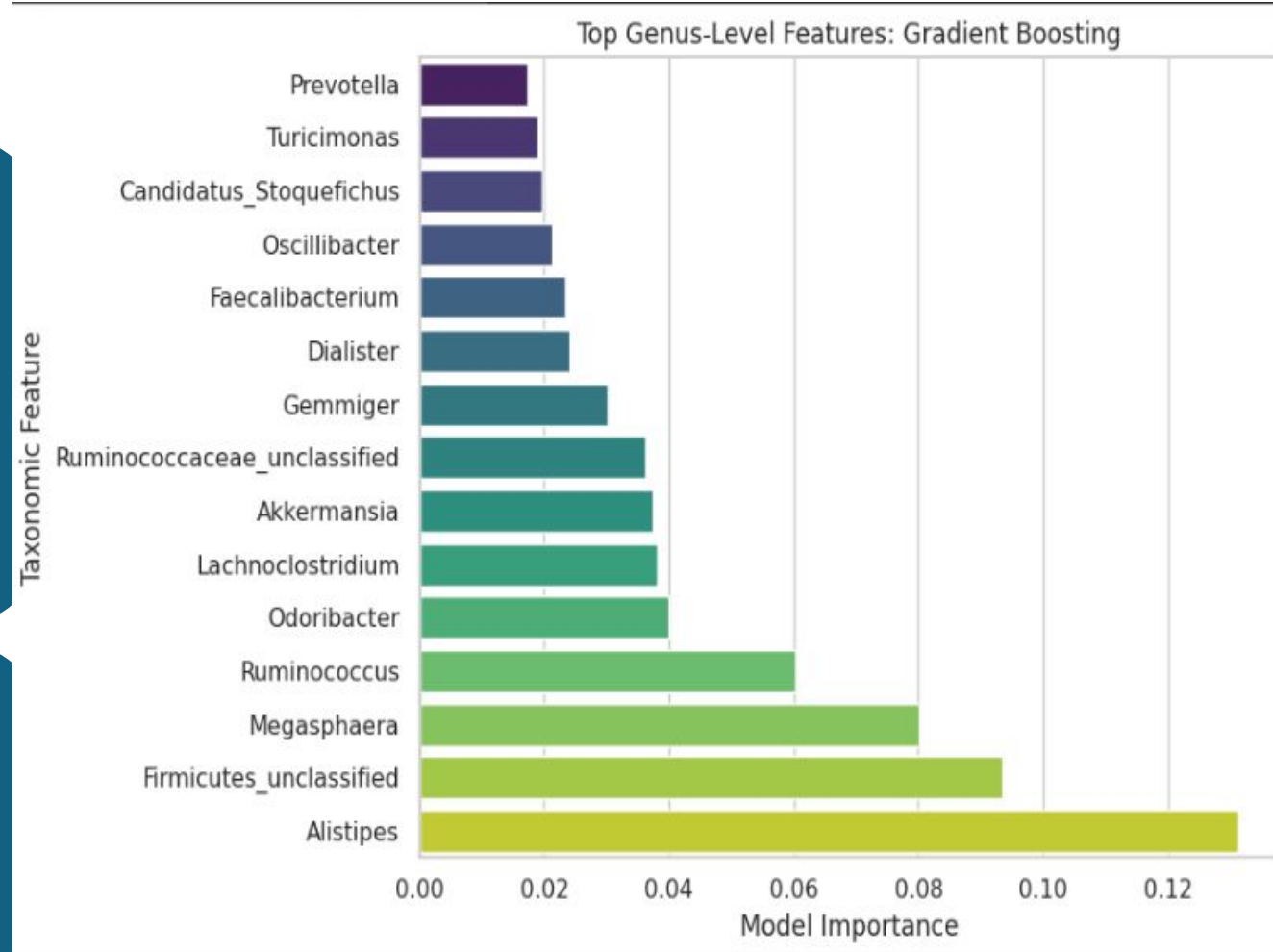


The precision-recall curve shows how precision changes as recall changes across thresholds for the IBD class.

Top Genus-Level Features: Gradient Boosting

Genera such as Alistipes, Firmicutes_unclassified, Megasphaera, and Ruminococcus were among the strongest predictive signals in this dataset

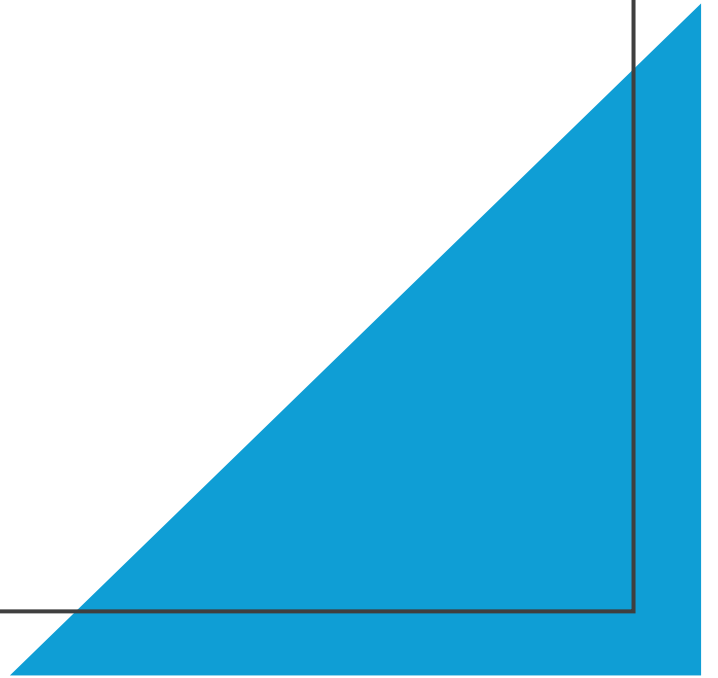
These are model-derived importance scores, so they support biological interpretation but do not prove causal relationships



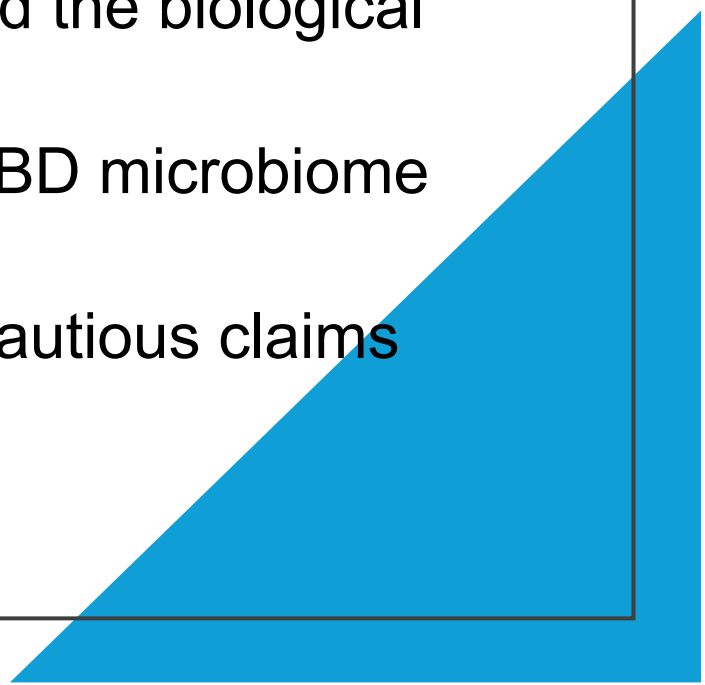
The feature importance chart identifies the genera that contributed most to the final Gradient Boosting predictions

Orange Results

- Orange was used as a supporting comparison environment for the exported final dataset
- The same binary target and selected microbial features were used to compare standard classifiers visually
- Orange results serve as an extra validation layer and presentation-friendly comparison, not the primary final model source



Literature connection

- Lloyd-Price et al. (2019) justified the HMP2 / IBDMDB dataset and showed strong IBD-associated microbial dysbiosis
 - Morgan et al. (2012) and Gevers et al. (2014) supported the biological plausibility of microbiome-based IBD classification
 - Manandhar et al. (2021) supported supervised ML for IBD microbiome prediction
 - Nguyen et al. (2022) motivated careful validation and cautious claims
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Conclusion and Future work

- Microbial taxonomic abundance contains useful predictive signal for IBD vs nonIBD classification
 - Gradient Boosting gave the strongest ranking performance in this pipeline
 - Threshold tuning improved the tradeoff between IBD recall and nonIBD specificity
 - This is a research-support screening model, not a clinical diagnostic tool
 - external validation, alternative transformations, additional feature types, and subject-level longitudinal modeling
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